

Thermal Analysis:- [TGA, DSC]

Temperature is a function of state variables

Thermal equilibrium:- $f(P, V_1) = f(P, V_2)$

simultaneously
called as temperature. $T_1 = T_2$

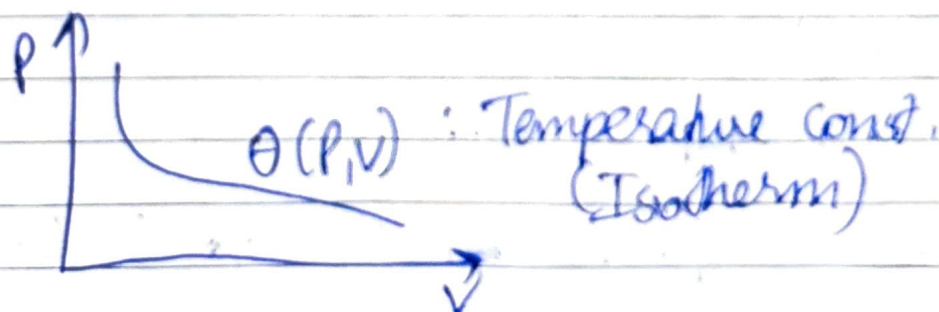
Black (1760)

→ Experiment by Black:- disproved fluid theory of temperature flowing from higher to lower.

→ $\Delta U = W$ {adiabatic system}
 $= Q + W$ {First Law of Thermodynamics}
 $dU = dQ + dW$

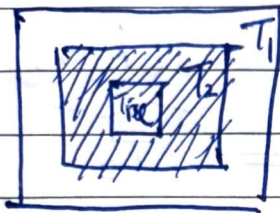
$H = U + PV$ {Total Energy = Enthalpy}

→ Temperature: follows from Zeroth Law of Thermodynamics.



→ 2 bodies in thermal equilibrium have the same temperature.

Black's Experiment:



Fluid flow of heat expects water to warm first ~~and~~ followed by ice to melt.

Observed:- ice melts ~~more~~ before water warms.

→ cannot be explained by fluid flow of heat.

Joule's heat:-

Heat Capacity:-

↳ heat energy required to make a small change in a state observable T

$$C^{(q)} = \frac{dQ}{dT} \Big|_{\text{other variables}}$$

$$C^{(T)} = \frac{\partial Q}{\partial T} \Big|_{V, P}$$

In hydrostatic pressure case: ~~dQ~~ $Q = U + p dV$

Calorimetry:-

↳ measurement of heat change after a process occurs.

→ Thermal Analysis:-

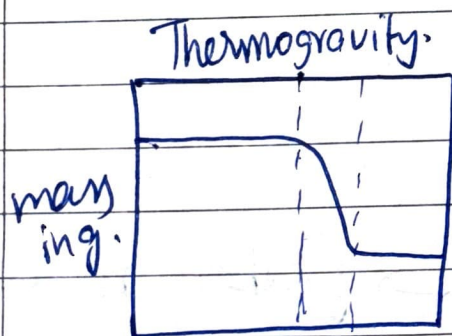
Thermal Transitions:-

↳ Physical phenomenon.

Chemical phenomenon:

Thermal Gravimetry:-

↳ mass change in substance measured as a fnc. of T whilst the substance is subjected to a controlled temperature program.

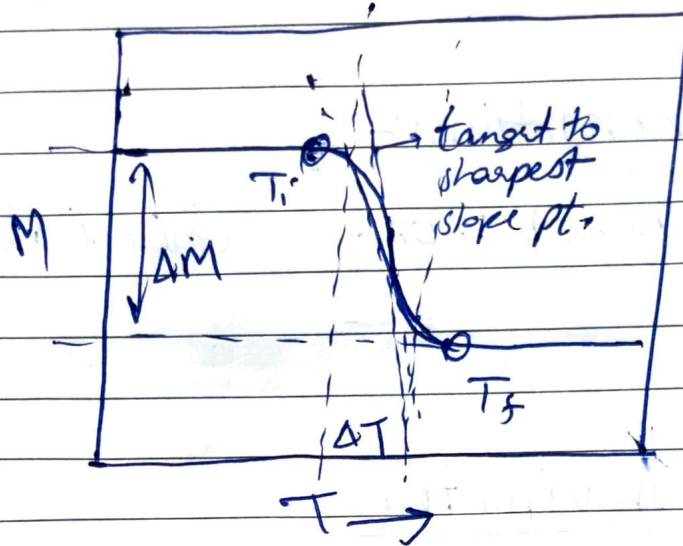


T_r → width of transition dictates type of thermal transition.

Microbalance:- works on quartz oscillators.

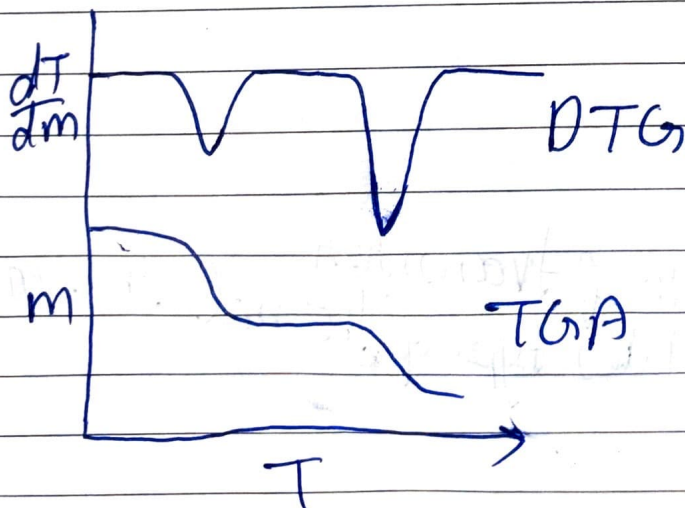
- Types of thermogravimeter:
- i) isothermal measurements
 - ii) scanning / ramping / rising measurements.

Analysing a measurement:-



Differential TG:-

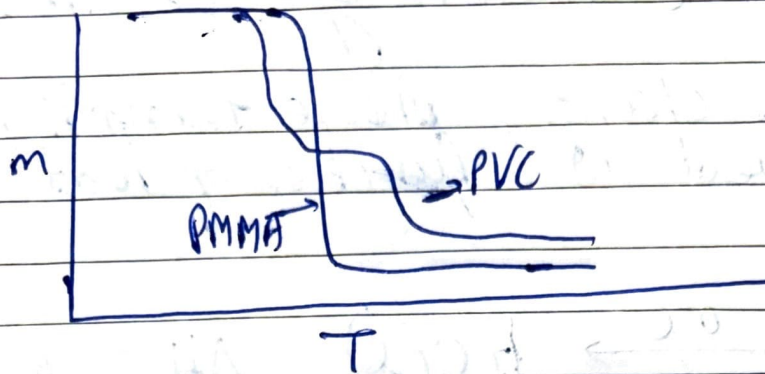
- ↳ easier to measure sharp transitions
- ↳ area under curve measures mass of substance in DTG.



→ peaks can be observed during mass gain.
eg. - Oxidation.

★ If you want to know the accurate temperature for drying / dehydration.

Example degradation of polymers:



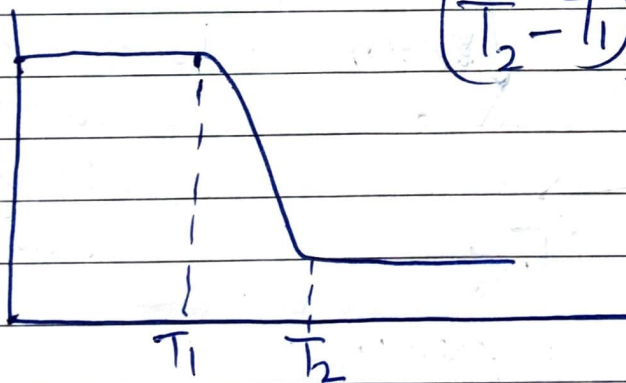
Factors affecting TGA measurement:

i) Heating rate:- T_1 (start) T_2 (end)

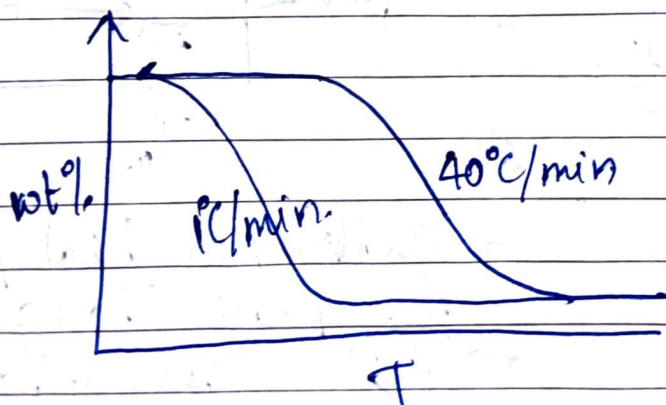
$$(T_1)_{\text{fast}} > (T_1)_{\text{slow}}$$

$$(T_2)_{\text{fast}} > (T_2)_{\text{slow}}$$

$$(T_2 - T_1)_f > (T_2 - T_1)_s$$

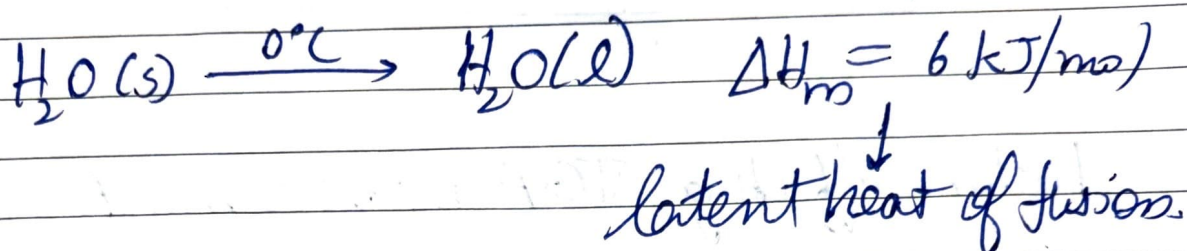


ii) Sample related:- Reaction is endothermic/exothermic.

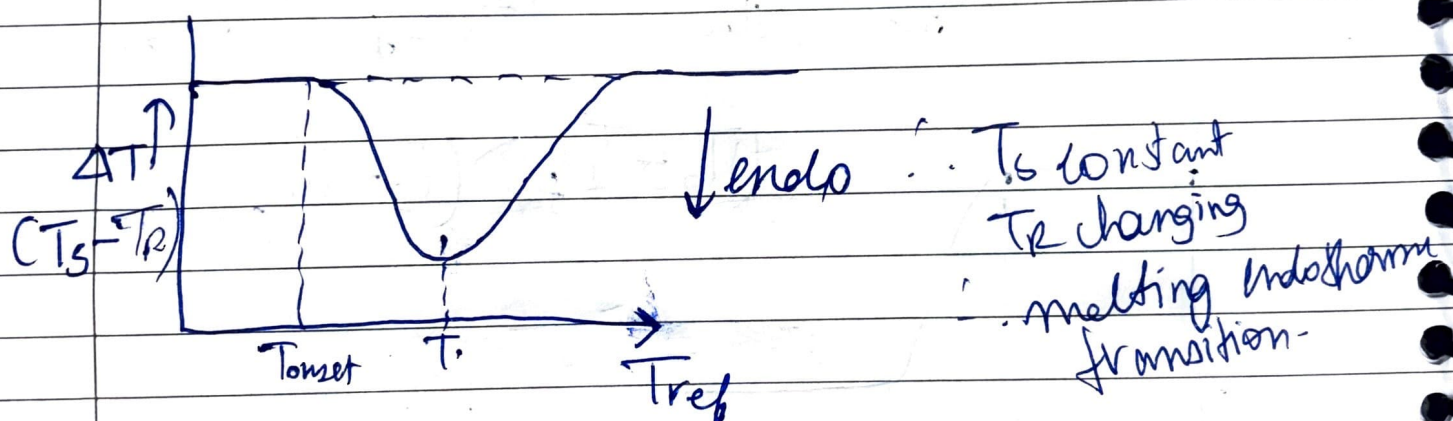


Differential Scanning Calorimetry:-

→ no mass change, phase transitions can be observed as a function of heat change.

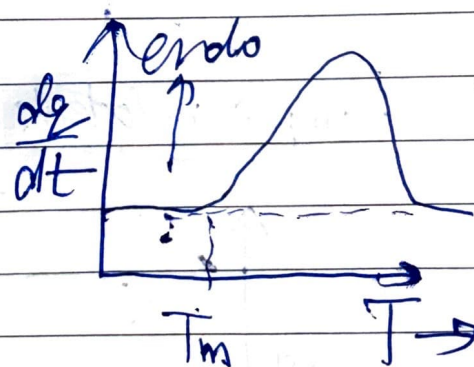


differential thermal analysis:-



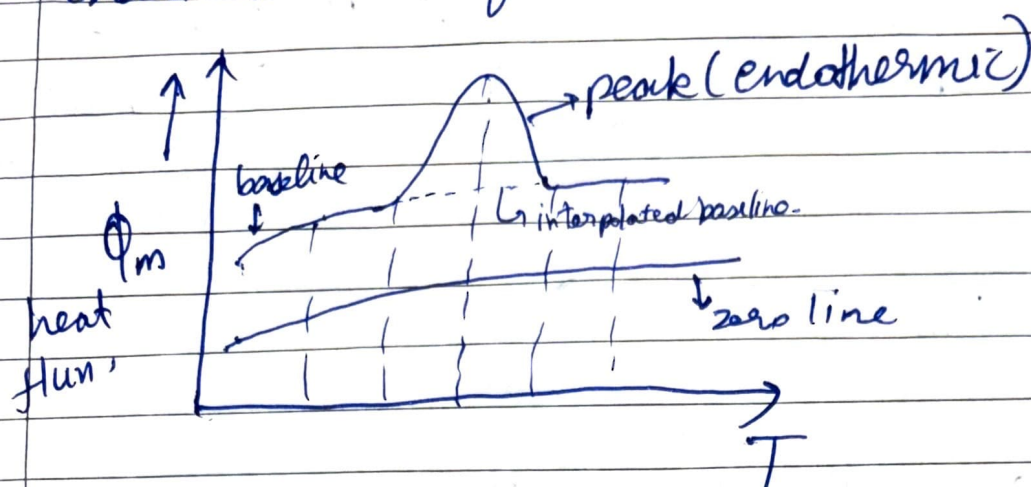
sample
S

reference.
R



→ additional heat provided for phase change.
→ this is observed as a hill in DSC.

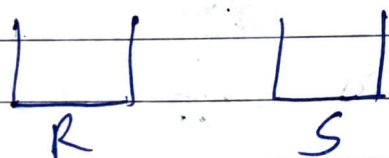
Characteristics of DSC curves:-



∴ Baseline - zero line $\Rightarrow$ Specific heat capacity.

Measured Enthalpy: area of peak: - $\Delta H = \frac{K M}{A}$

DTA



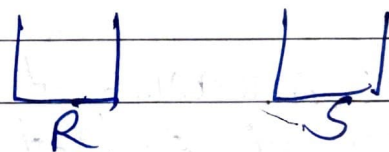
$$\frac{dQ}{dt} = \text{const.}$$

$\rightarrow$ change in temperature measured;

$\rightarrow \Delta T \text{ vs } T$

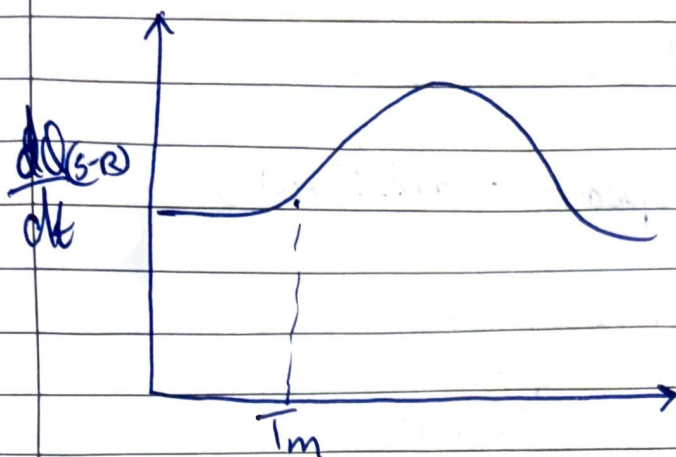
$(T_s - T_R)$ vs T_R provided reference material does not undergo any phase transition.

DSC



$$T = \text{const.} \therefore \underline{\Delta T = 0}$$

$\Rightarrow$ change in heat measured.



In DSC: $\Delta T = 0$
 $T_s = T_r$ at any time t .

The Ohm's law for heat:

$$\Delta T = R * \frac{dq}{dt}$$

R = thermal resistance $\rightarrow q$ = heat supplied.

In DTA: $\frac{dq}{dt} = C_s \frac{dT_s}{dt}$

$$\therefore \Delta T = T_s - T_r = R(C_s - C_r) \frac{dT}{dt}$$

The measured temperature difference depends on heat capacity.

Power compensated DSC: -

$$T_s - T_s^0 = \Delta T_s = R * \left(\frac{dq_s}{dt} \right)$$

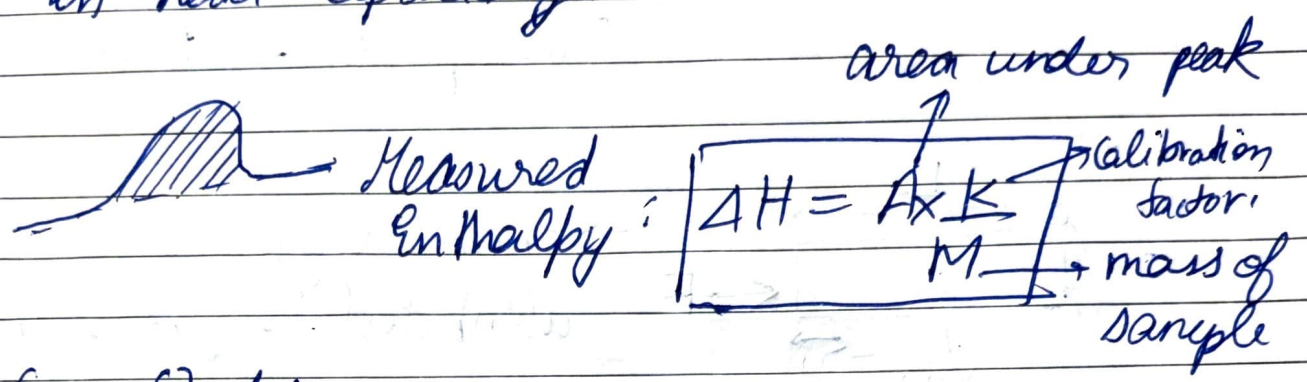
$$T_r - T_r^0 = \Delta T_r = R * \left(\frac{dq_r}{dt} \right)$$

$$\frac{dq}{dt} = C \frac{dT}{dt}$$

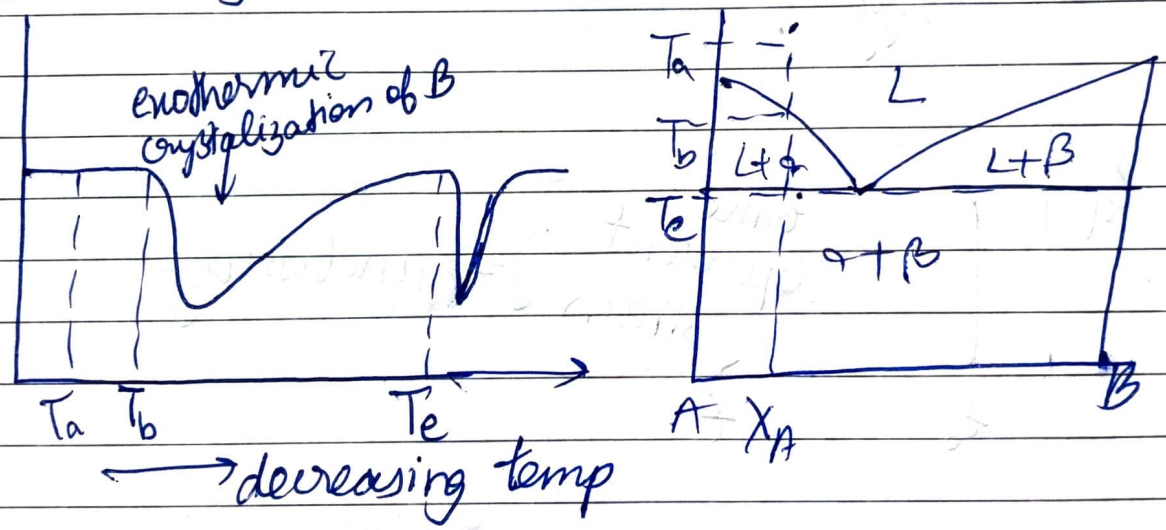
$$\Delta \frac{dq}{dt} = \frac{dT}{dt} (C_s - C_l)$$

→ no dependence on thermal resistance of system

Phase transition is associated with a change in heat capacity.



Case Study:



DSC of polymeric samples:-

